

Luigui Gallardo-Becerra

SOFTWARE ENGINEER & DATA SCIENTIST

☎ +1 619 602 0725 | ✉ luiguimichelgallardo@gmail.com | 🌐 www.lmgb.xyz | 📷 LuiguiGallardo | 📺 luiguigallardo

Software Engineer with 6+ years of experience in designing scalable data systems, automating complex workflows, and building cloud-native solutions. Proficient in Python, SQL, and DevOps practices, with a strong background in optimizing high-performance computing environments and backend development.

Experience

Graduate Research Assistant (Data Science & Bioinformatics)

Institute of Biotechnology, UNAM

January 2019 - July 2025

- Engineered automated, containerized pipelines using Nextflow, Docker, and Snakemake for reproducible data processing.
- Optimized distributed computing systems and high-performance computing (HPC) workflows for large-scale analysis.
- Developed machine learning models to analyze complex datasets, resulting in 14 peer-reviewed publications.
- Conducted statistical analyses and feature engineering to identify trends and patterns in biological data.

Data Engineer

Appen

October 2016 - December 2018

- Built robust SQL-based ETL architectures for data integration and automated reporting systems.
- Developed and automated data collection workflows using Python and Bash, reducing processing time by 50%.
- Preprocessed and analyzed large-scale datasets for AI-driven projects, improving data quality by 30%.

Software Engineer (Internship)

Dept. of Computer Science, CUCEI

January 2016 - July 2016

- Developed a cloud-deployed web application using ASP.NET Core for patient record management, contributing to both backend and frontend development.

Education

National Autonomous University of Mexico (UNAM)

PH.D. IN COMPUTATIONAL BIOLOGY

Mexico City, Mexico

January 2019 - July 2025

National Autonomous University of Mexico (UNAM)

MASTER OF SCIENCE IN COMPUTATIONAL BIOLOGY

Mexico City, Mexico

August 2016 - January 2019

University of Guadalajara (UDG)

BACHELOR OF SCIENCE IN MOLECULAR BIOLOGY

Guadalajara, Mexico

August 2012 - January 2016

Key Skills

- **Programming & Frameworks:** Python, SQL, Bash, C#, JavaScript, Django, React, R
- **Cloud & Infrastructure:** AWS, Docker, HPC, CI/CD, Linux, Database Management
- **Data Engineering:** ETL Pipelines, Apache Spark, System Automation, Large-Scale Data Processing
- **Tools:** Git, VS Code, Jupyter, Snakemake, Nextflow
- **Soft Skills:** Problem-solving, Technical Leadership, Project Management, Agile Collaboration
- **Languages:** English (Full professional proficiency), Spanish (native)